

JUNIOR PHILIPPINE SOCIETY OF MECHANICAL ENGINEERS

How the system works

Every screen, every flow, and every state a screen has to handle.

A reference for designing screens for the JPSME membership and events platform. Written for somebody who has to draw these screens without reading the code: what each screen is for, who sees it, what it shows, and — the part usually missing — every state it can be in.

SECTION 01

The shape of it

Three kinds of person use this system, and they see almost nothing in common. Designing for the wrong one is the easiest mistake to make here, so this is first.

Member / visitor	The public site. Dark navy, gold accents, marketing-weight pages plus their own profile, tickets and seat. Mostly read on a phone.
Chapter admin	The admin panel, light theme, but only their own chapter's members — and event-day stations at events they have been granted.
Main admin	Everything: members, events, payments, certificates, settings, the audit log.

The two themes are not one theme The public site is dark (#080f29 ground, white text, gold accents). The admin panel is light (white cards, slate text, indigo accents). They share a logo and nothing else. Do not design one and assume the other.

The four things the system does

- Membership — somebody signs up, verifies their email, pays a fee, and an admin approves them.
- Events — published events people register for, some of them with a fee.
- Event day — a registration desk and hall doors, both driven by QR scans.
- Records — certificates, reports, broadcast email, and an audit log of who did what.

SECTION 02

Flow: becoming a member



The verification step is the one with the most states, and the one most often drawn wrong. The code is sent automatically — the member never asks for it — and it expires in three minutes, shown as a draining ring rather than a number alone.

States the verification screen must cover

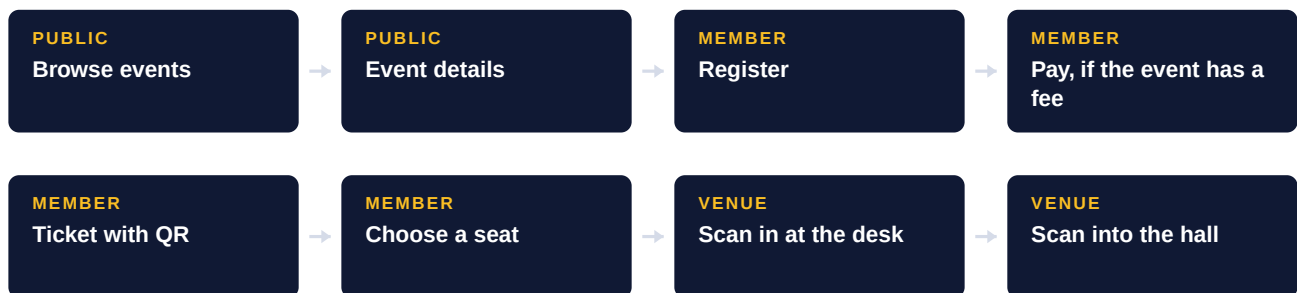
Waiting	Six empty digit boxes, the ring full, the code already sent.
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Counting down	The ring drains over three minutes. Below it: "Didn't get the code? Resend" with a 60-second cooldown counting down before Resend becomes clickable again.
Expired	The ring is empty. The only way forward is Resend.
Wrong code	An error, and the digits they typed stay where they are.
Human check failed	A new challenge image, and — importantly — the typed digits survive.
Verified	Straight to sign-in with the email already filled in.

Resend needs no human check Verifying does. The difference matters: a resend only re-sends a code to an address that is already on the account, so putting a puzzle in front of it is friction that protects nothing.

SECTION 03

Flow: attending an event



The ticket

One screen a member will actually hold up at a door, so it is built for that: a QR on a deliberately white card even though the site is dark, because a scanner reads dark-on-light and an inverted code will not scan. Below the card, off it: their seat, their movements, and a button to choose or change a seat when the event uses assigned seating.

Choosing a seat

A plan of the hall the member taps directly. Taking a seat is two steps, because the plan is being changed by other people while they look at it:

1. Hold	Tap a free seat and it is held for five minutes, with the countdown on screen.
2. Confirm	Confirm makes it theirs. Give up releases it immediately.
If they lose it	"Somebody just took that seat. Please pick another." This is normal, not an error, and must not be drawn as a failure.

An attendee never sees another attendee's name On the member seat plan, held / assigned / occupied / stepped-out all look identical — simply "taken". Only their own seat is distinct. Which seat belongs to whom, and whether that person is currently in the room, is staff information.

SECTION 04

Flow: the event day

Two stations, and keeping them apart is the whole design. There was once a third — a venue entrance — and it was removed: for a single-hall event it asked a question the hall door had already answered.



Registration desk

Scan a ticket to LOOK SOMEBODY UP — it admits nobody by itself. Staff then give them a seat, and only then check them in. Happens once per person.

Hall door

In and out of the hall, as often as somebody comes and goes. One scan does both: the server decides the direction from where the person currently is.

One button, not two The hall door has a single scan action. An operator with a queue in front of them must not have to choose between "in" and "out" — so the screen shows the direction as the RESULT of a scan, in the largest type on the page.

What a scan screen must show

- A verdict readable from arm's length — the operator is standing, holding a scanner.
- The person's name and registration number, large.
- Colour that separates came-in from went-out from refused, at a glance.
- A live count of how many people are in the hall.
- Nothing that needs a mouse. The screen refocuses its own input after every scan.

Scan results a screen has to handle

 ENTERED	Came into the hall. Green.
 LEFT	Went out. Blue — not green, and not red.
 ALREADY SCANNED	The gun fired twice, or the operator rescanned. Nothing is wrong and nothing changed. Amber.
 ROOM FULL	The hall is at capacity.
 REFUSED	Not registered, cancelled, unpaid, wrong event, or an unreadable code. Always says which.

SECTION 05

The seat lifecycle

The part of the system with the most states, and the one a designer most needs a reference for. A seat moves through these on its own as people scan in and out of the hall.



The six states, and the colours the admin plan uses

AVAILABLE	Nobody's. White with a grey border on the admin plan.
 HELD	Somebody is choosing it right now. Expires after five minutes.
 ASSIGNED	Theirs, but they have not been seen in the hall yet.
 OCCUPIED	Theirs, and they are in the hall now.
 AWAY	Theirs, but they have walked out. Still theirs — for ten minutes.
 BLOCKED	Not in use. Struck through.

AWAY is the state the whole thing exists for It is a seat that looks empty from across the hall and is not free. Ten minutes after somebody leaves, their seat is released automatically and can be given to somebody else. Staff also get a "Stepped out" list — who left, how long ago, how long until the seat frees — with a Free button for when they know that person has gone home.

SECTION 06

Every screen

Public and member

/	Home. Hero, upcoming events, sponsors, stats.
/about/...	Five pages: the society, quality policy, code of ethics, theme of the year, officers.
/events	Event list.
/events/:id	Event details, and the Register button in all its states.
/events/:id/ticket	Their ticket: QR, seat, movements.
/events/:id/seat	The seat plan they pick from.
/articles, /articles/:id	News.
/chapters, /organizations	The society's structure, browsable.

/register, /login, /verify-email	Sign up, sign in, verify.
/profile	Their membership, registrations, tickets, certificates.
/membership-payment	GCash checkout for the membership fee.

Admin — event day

/admin/check-in	Which events this account can work.
.../attendance	Event day hub: the funnel, and a way into each station.
.../desk	Registration desk: look up, seat, admit.
.../rooms	The halls, with live occupancy.
.../rooms/:id/scan	The hall door scanner.
.../seating	The seat plan, the stepped-out list, and the build tools.
.../check-in/report	Every scan that happened, including refusals.

Admin — everything else

Members	Users, approvals, per-chapter member lists, organisation admins.
Events	Create, edit, registrations, invitations, per-event email.
Money	Payments, refunds, the membership fee setting.
Records	Certificates, broadcast email, the audit log.
Content	Articles, sponsors, site settings.

SECTION 07

Statuses, as the system names them

These are the exact values the system stores. A design that invents a status outside these lists produces a screen nobody can build.

Account status	PENDING · APPROVED · REJECTED
Role	USER · CHAPTER_ADMIN · ADMIN
Event registration	REGISTERED · PENDING_PAYMENT · CANCELLED
Payment	PENDING · PROCESSING · PAID · FAILED · EXPIRED · CANCELLED · REFUNDED
Seat	AVAILABLE · HELD · ASSIGNED · OCCUPIED · AWAY · BLOCKED
Seat type	REGULAR · VIP · ACCESSIBLE · TABLE

In the hall

INSIDE · OUTSIDE

Scan result

SUCCESS · ALREADY_CHECKED_IN · DUPLICATE_SCAN · INVALID_QR ·
WRONG_EVENT · NOT_REGISTERED · CANCELLED · UNPAID · REJECTED ·
ROOM_FULL · ROOM_CLOSED · UNDONE

SECTION 08

Things that constrain the design

Not preferences. These are properties of where this runs, and a design that ignores them cannot be built as drawn.

The public palette, as it actually is

	Page ground	#080f29 — the darkest surface, behind everything.
	Card	#111b3a — panels and cards on the public site.
	Deep panel	#0a1128 — insets and tinted blocks.
	Accent	#fbbf24 — the one accent. Links, emphasis, active states.
	Body text	#94a3bd — secondary text on dark. Headings are white.
	Border	#263b70 — separators on dark.

Constraints worth knowing before drawing

- No external fonts or scripts. The site loads nothing from a CDN, so a design that depends on a hosted webfont needs that font embedded instead.
- Uploaded images are capped at 5 MB and stored in the database in pieces. Avoid designs needing many large images per page.
- The QR on a ticket must stay dark-on-light with a quiet zone around it, whatever the surrounding theme. This is a scanning requirement, not a style choice.
- Scanner screens are used standing up, one-handed, at arm's length, sometimes in a noisy hall. Large type, strong colour, no small targets.
- Every list can be empty on a fresh deployment. Empty states need designing, not leaving blank.
- Tables on phones must scroll inside their own container — the page itself must never scroll sideways.

If you change one thing, make it the empty and error states Most screens here are drawn well for the happy path and thinly for everything else. The refusal at a door, the seat somebody just lost, the expired code, the event with no registrations yet — these are the screens staff and members actually remember.

